

Process Performance Metrics

U4 – Measure

E4 – Process Capability and Performance

The element 'Process Capability and Performance' explains process capability and performance in relation to specification limits.

The importance of metrics

- Two important management tasks: sustain and improve
- We must make performances measurable so that we can ensure that performance will not deteriorate
- We must make performances measurable so that we can make the results of improvement activities visible

Lean Six Sigma metrics

Metrics = Performance Indicator

Lean & Six Sigma both have their own performance metrics - each with a different focus:

- **Lean metrics focus on effectiveness and (delivery) time**
- **Six Sigma metrics focus on quality and capability**
- **Both focus on costs costs (€) and efficiency (%)**

Takt time

- Takt time: the average time between two consecutive customer procedures/ processes/ consultations
- 'Takt' is German and means the pace determined by the orchestra's conductor
- The customer's demand determines the Takt time (not hospital/ clinic / general practitioner)

$$\text{Takt time} = \frac{\text{Available production(process) time}}{\text{Required production (process) volume}}$$



Example Takt time in elderly home

Customer demand: 200 meals to be served at all private rooms between 17.00 and 19.00 every day

Available production (process) time:

- 2 hours per day
- 3 delivery employees

$$\text{Takt time} = \frac{\text{Available production(process) time}}{\text{Required production (process) volume}}$$

$$\text{Takt time} = \frac{3 \times (2 \text{ hours})}{(200 \text{ meals per } 2 \text{ hours per day})} = \frac{360 \text{ (minutes/day)}}{200 \text{ (meals/ day)}} = 1,8 \text{ min/meal}$$

Cycle time (C/T) & Lead time (L/T)

Cycle time

- The time between the completion of 2 successive procedures/ processes/ tasks by observation
- The time it takes personnel to go through all of the work elements before repeating them

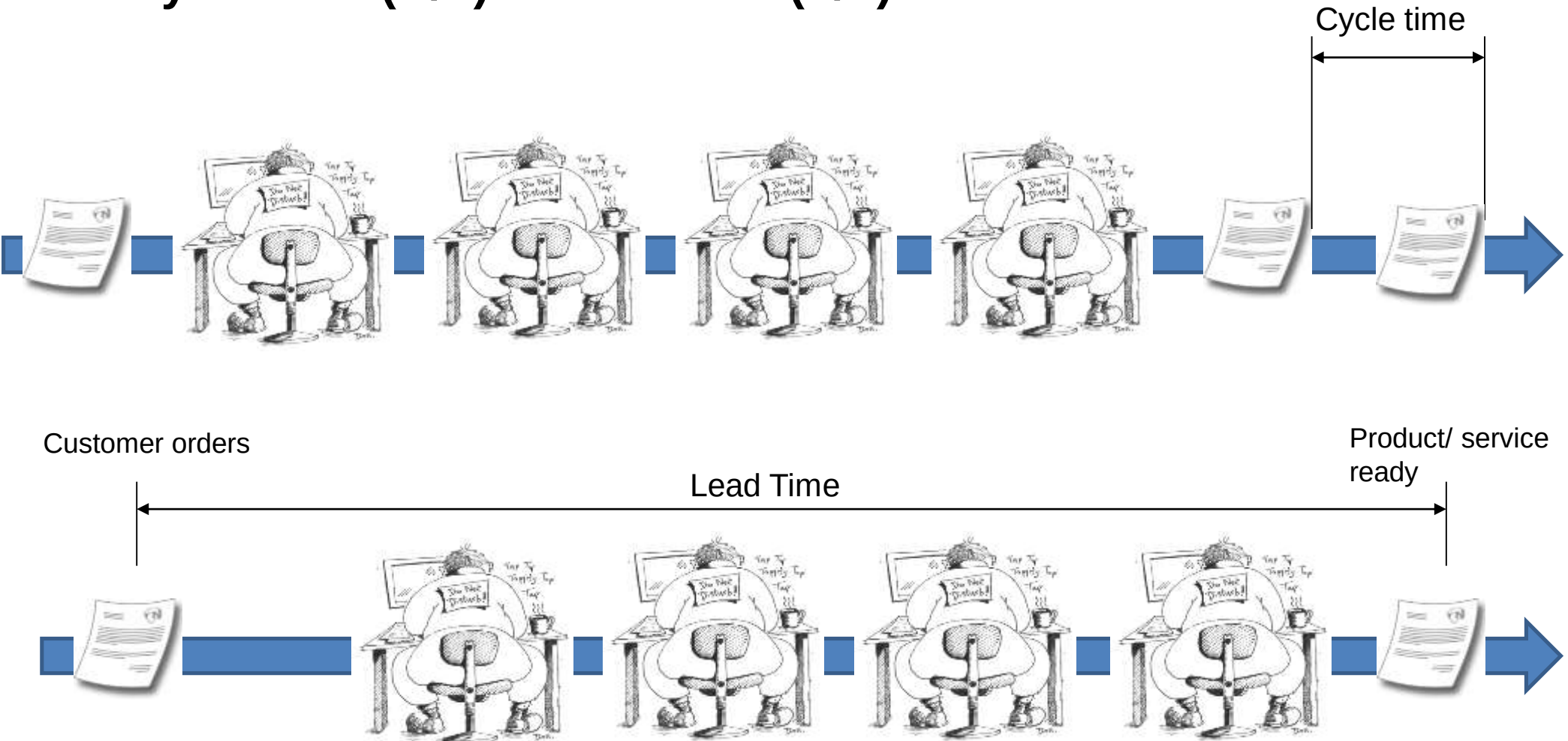
Lead time

- Time needed for ONE part to pass the complete process/ value stream, from start to finish
- Example: a nurse can take a patient for a blood sample in runs over a day every 10 minutes, but the actual (value adding) process (taking the blood) takes only 30 seconds:

Value Added Time	= 30 s
NON Value Added Time	= 9 min 30 s
Lead Time	= 10 minutes

Process Performance Metrics

Cycle time (C/T) & Lead time (L/T)



Defects per unit (DPU)

DPU is the number of defects per unit

- DPU can be greater than 1 because a unit can have several defects (for instance: a prescription may have more than 1 error)

Example

- Number of units = 3000 number of written prescriptions
- Number of defects = 53 number of wrong or missing dosages

$$\text{DPU} = \frac{\text{number of defects}}{\text{number of units}} = \frac{53}{3000} = 0.0177$$



- So per prescription received into pharmacy, there are on average 0,02 defects

Defects per million opportunities (DPMO)

- Opportunities = 2 possible defects per unit: missing & wrong
- Number of units = 3000 number of prescriptions issued
- Number of defects = 53 number of missing and wrong dosages

$$\text{DPMO} = \frac{1.000.000 \times \text{Number of Defects}}{\text{Opportunities} \times \text{Number of Units}} = \frac{53 \times 10^6}{2 \times 3000} = 8.833$$



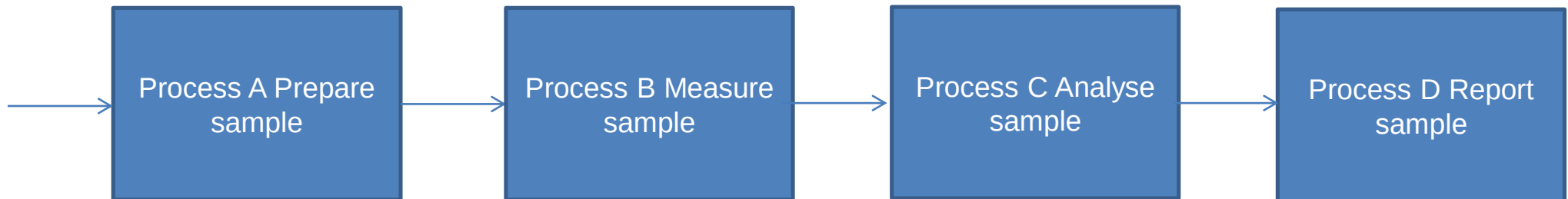
- In other words: if we write 500.000 prescriptions, we can expect that there will be 8.833 defects (missing and wrong dosages)

Yield [%]

- **Process yield** =
$$\frac{\text{Total number of processes/procedures without defects}}{\text{Total number of processes/procedures}}$$
- **First time right (FTR) yield** =
$$\frac{\text{Total number of processes/procedures right first time}}{\text{Total number of processes/procedures}}$$
- **Rolled throughput yield**
The probability that a product/ service will pass a number of sequential processes-steps without any defects (first time right)

Example Process yield

Process of testing bloodsamples in a local laboratory. When patients visit a general practioner sometimes a blood sample is taken. This blood sample is sent to the local laboratory to be analysed.



Process Performance Metrics

Process yield

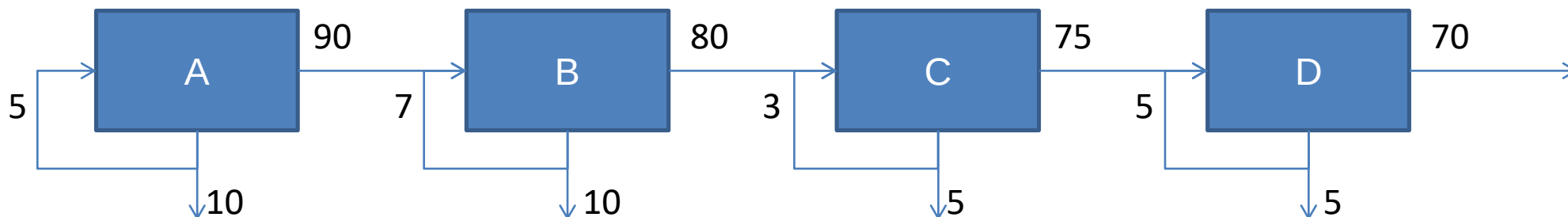
Process	Scrap	Out	Process yield		
A	10	90	$90 / (90+10)$	$= 90 / 100$	$= 90\%$
B	10	80			
C	5	75			
D	5	70			

- Process yield =

Example Rolled throughput yield

In the local laboratory some samples have to be reworked

- Process A : 100 blood samples in; 10 scrapped + 5 reworked; 90 out
- Process B : 90 blood samples in; 10 scrapped + 7 reworked; 80 out
- Process C : 80 blood samples in; 5 scrapped + 3 reworked; 75 out
- Process D : 75 blood samples in; 5 scrapped + 5 reworked; 70 out



Process Performance Metrics

First time right yield and Rolled throughput yield

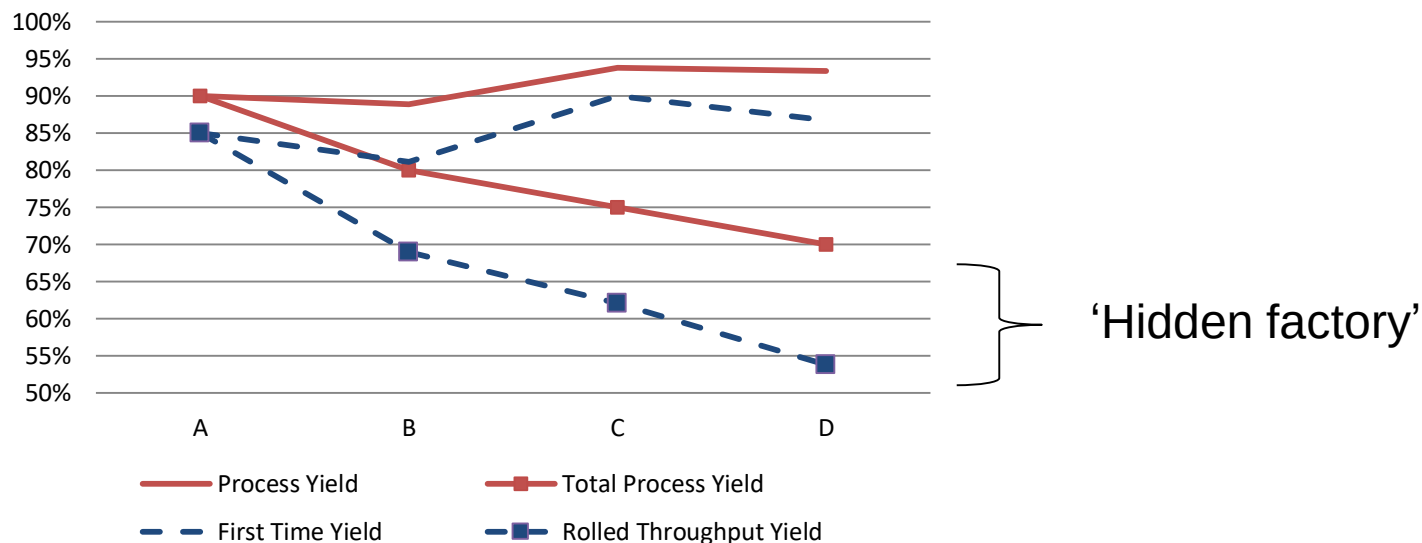
Process	Scrap	Rework	Out	First time right yield		
A	10	5	90	$(90-5) / (90+10)$	= 85 / 100	= 85%
B	10	7	80			
C	5	3	75			
D	5	5	70			

- Rolled throughput yield =

Process Performance Metrics

Summary Yield

Proces	Units out	Scrap	Rework		Process Yield	Total Process Yield		First Time Right Yield	Rolled Throughput Yield
A	90	10	5		90%	90%		85%	85%
B	80	10	7		89%	80%		81%	69%
C	75	5	3		94%	75%		90%	62%
D	70	5	5		93%	70%		87%	54%



Process Capability & Performance

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E4 – Process Capability and Performance

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Definition of process capability:

‘The ability of a process to supply products/ services that satisfy customer expectations / specifications’

Example Elderly home centre

- The specifications of the ideal temperature of a meal in an Elderly home centre are: between 60 and 70 degrees
- A capability analysis indicates how capable the process is to meet specifications and how the process can be improved and secured
- In order to reliably determine the process capability, the process must be stable
- An unstable process is unpredictable

Example Temperature problems

- As a result of a survey on the meal service of an Elderly home centre the temperature of the meal was found to be an issue with clients. E.g. too hot or too cold
- Interviews learned that the ideal temp was between 60 and 70°C



Lower specification limit
(LSL)

Upper specification limit
(USL)

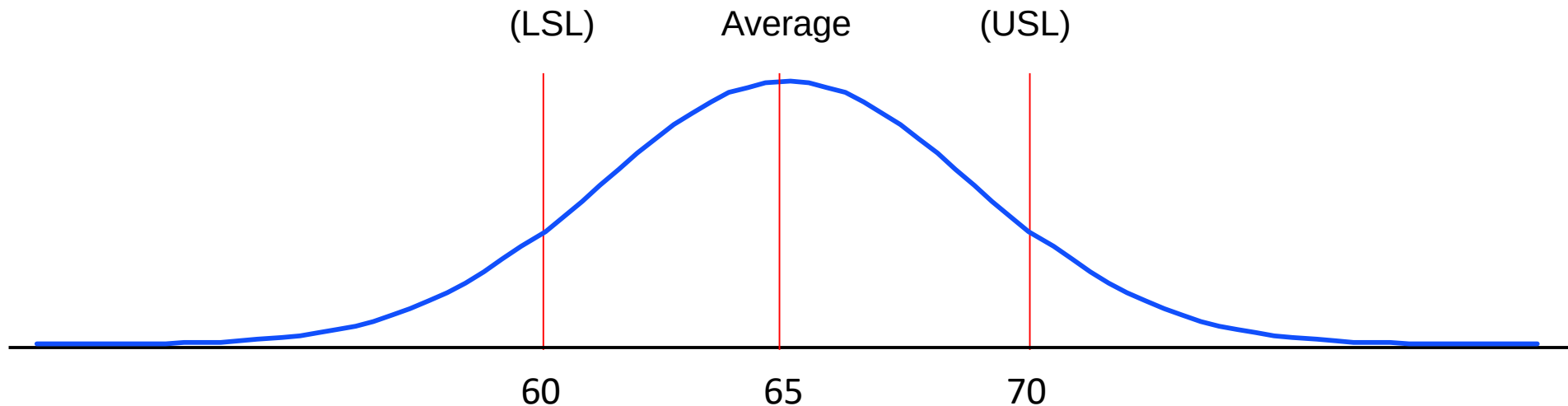
60

70

Example Temperature problems

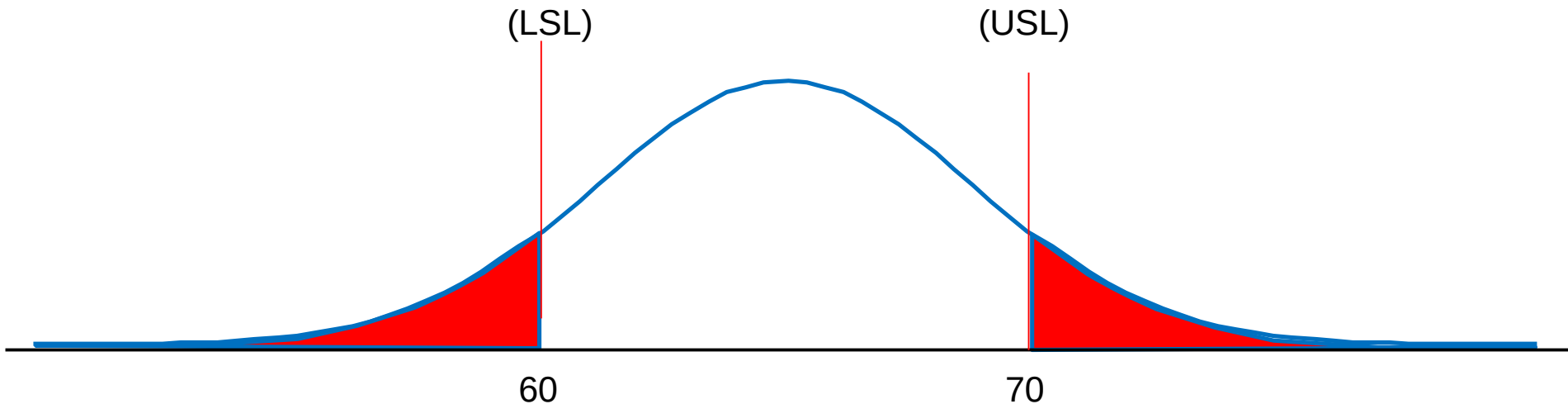
The Elderly home centre started to measure the temperature of the meals

- The average temperature was 65°C
- The standard deviation was 10°C



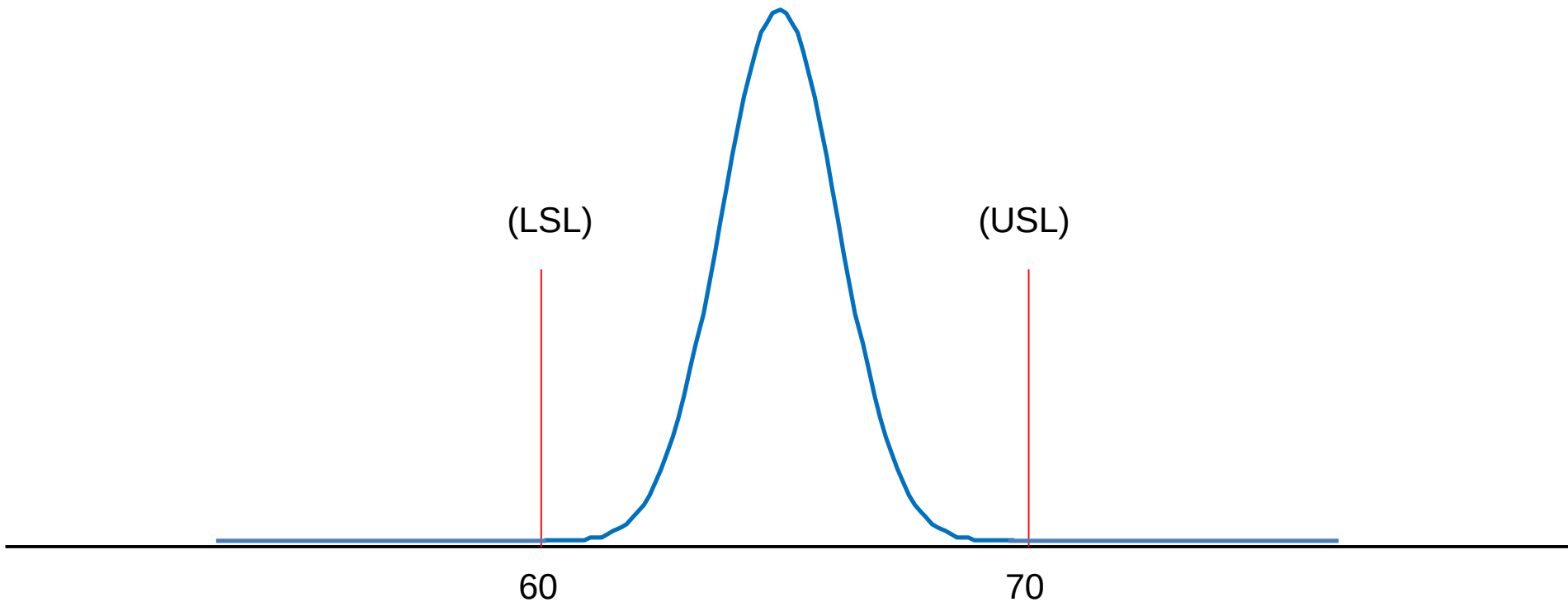
Example Temperature problems

Based on the information collected we can see that approx. 30% of the meals served is either too hot or too cold



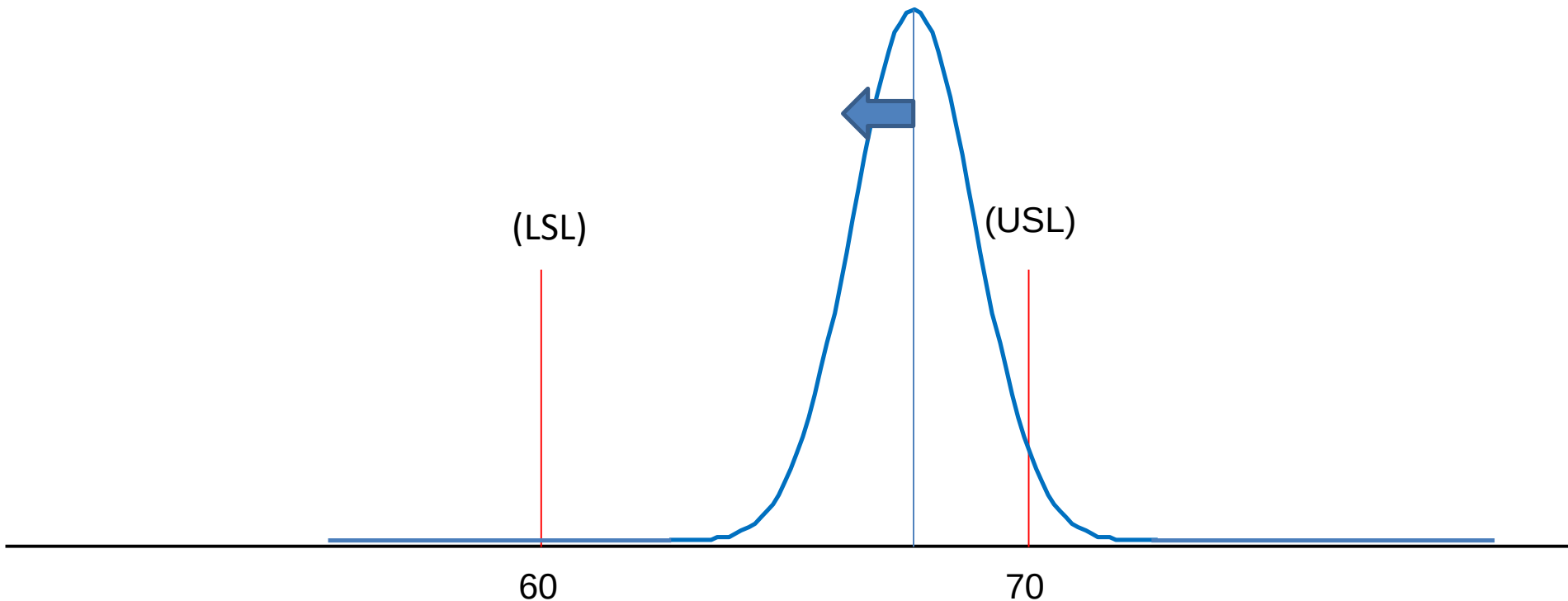
Example Temperature problems

So, in order to meet clients expectations, they had to improve the process by reducing the variation in the meal temperature



Example Temperature problems

Sometimes it's not (only) the variation, but (also) the average that needs adjusting



Capability vs. Performance

- **Process capability:** The potential capability of the process (Cp and Cpk)
- **Process performance:** The processes overall performance (Pp and Ppk)
- **What the process can do (Cp) and what the process does (Ppk)**

Short term vs. long term capability

